Knowledge Management Systems and Technologies

Activity 3

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Introduction

In an organization, useful knowledge is usually spread across policies, incident tickets, FAQs, and meeting minutes, and employees spend time looking for a document that was already written or solving a problem that was already solved. A recommender system addresses this situation by suggesting related documents from a seed document or from a short text. This report describes a Top-K recommender built with TF-IDF and cosine similarity, which was validated with three tests on a simulated corpus of 31 documents. It presents the approach, the results of the tests, and the areas of opportunity identified, and it is accompanied by the source code (recommender.ipynb), the dataset (corpus.csv), and the evidence of the tests.

Description of the approach

This recommender suggests the five documents most similar to a seed document or to a free-text query, so that employees can reach related policies, incident tickets, FAQs, and meeting minutes without searching each repository separately. Path A (TF-IDF) was selected because it is simple and transparent: every score can be traced to words that two documents share.

The corpus (corpus.csv) contains 31 simulated documents with the fields id, title, text, and category: 11 Policy/Process, 10 Support/Incidents, 5 FAQ, and 5 Meeting minutes. The title and content of each document are cleaned by converting to lowercase, removing accents and special characters, removing English stopwords, and applying light stemming (the first six letters), so that “password” and “passwords” count as the same term. The cleaned text is vectorized with TfidfVectorizer, and cosine similarity compares the seed or query vector with the whole corpus. When an ID is used, the seed document is excluded from the results. The notebook (recommender.ipynb) also validates invalid IDs, empty queries, queries with no known terms, duplicated identifiers, and missing files.

Test results

Three tests were executed to validate the recommender: a seed document from Policy/Process, a seed document from Support/Incidents, and a free-text query. The following table summarizes the input of each test, the best-ranked recommendation with its similarity score, and a brief assessment of the coherence of the Top-5.

Test	Input	Best result (score)	Assessment
1. Policy/Process	DOC001 Corporate Password Policy	DOC022 FAQ: How do I reset my password? (0.193)	Top 3 coherent; rank 4 (DOC005, vacations) is weak
2. Support/Incidents	DOC014 VPN failure for remote workers	DOC009 Remote Work and VPN Access Policy (0.324)	Top 2 very coherent; ranks 3-5 related but weaker
3. Free text	“I can't log in because I forgot my password”	DOC022 FAQ: How do I reset my password? (0.305)	Top 3 useful; ranks 4-5 weak (below 0.12)

The recommendations are coherent when documents share specific vocabulary, as in the VPN case, where the system returns both the governing policy and the end-user guide. Scores are moderate (0.07 to 0.32) because the documents are short, so the

relative ranking matters more than the absolute value. The complete outputs and comments are in the notebook and in the evidencias folder.

Areas of opportunity

The tests show that the recommender works well when documents share specific vocabulary, but they also reveal limitations that come from the technique and from the corpus. The following areas of opportunity are based on the results observed in the three tests and describe the improvements that would be required to use the system in a real environment.

- ***Synonyms and semantics:*** TF-IDF does not relate "log in" with "connect" or "access". Embeddings (Path B) would capture meaning.
- ***Ambiguity and short documents:*** generic words produce weak matches (DOC005 in Test 1). A minimum score threshold would avoid them.
- ***Corpus bias: topics*** with more documents, such as security and access, dominate the results.
- ***Metadata and role-based filters:*** department, date, and version would allow filtering by user role and prioritizing current policies.
- ***Security:*** per-document access control is needed so that confidential content is not recommended to unauthorized users.
- ***Continuous evaluation:*** user feedback and precision@5 on labeled queries would guide re-indexing when the corpus changes.

Organizational Reflection Video

The video, with a duration of 3 minutes, was recorded as a screen capture of the notebook with voice narration. It runs one of the tests (DOC014) and answers the three questions of the activity: how the recommender reduces rework and accelerates decision-making, which limitations were observed, and which improvements a real-world environment would require.

https://youtu.be/bsUbhFvphv0

Conclusion

The recommender fulfills its purpose, since it returns five ranked documents with similarity scores for both a document ID and a free-text query, and its top results were coherent in the three tests: for example, the VPN incident led to the remote work policy and the connection FAQ. Its main strength is transparency, because each score can be traced to shared words. Its main weakness is its dependence on exact vocabulary, so synonyms and generic words reduce the quality of the lower-ranked results. For these reasons, the approach is suitable as a first version that reduces rework and accelerates decision-making, while a real environment would require embeddings, metadata, role-based filters, security controls, and continuous evaluation, as described in the areas of opportunity.

The source code, the dataset, and the evidence of the tests are available in the following repository: ***https://github.com/FridaGarzaG/topk-document-recommender.git***